

CMPT 225-D100 Quiz 1, Summer 2026

Duration	30 minutes for the entire exam.
Coding Questions	For coding questions, use C++-like pseudocode. This means that small coding errors will usually be overlooked. However, it must be clear that the code could easily be converted to real C++.
Aids allowed	Pencil or pen, and eraser. No notes, no papers, no books, no computers, no calculators, no cheat sheets, etc.
During the exam	Raise your hand if you would like to speak with a proctor. To be fair to all students, we don't answer questions about the exam. Choose the answer based on your best understanding of the question. Re-reading the question may help.

Quiz 1

Consider a **doubly-linked list** (with Node defined as shown). Assume that H is a global variable of type Node* that is either nullptr or that points to the first node of the list. As usual, each node points to the next node and the previous node, and if there is no next/previous node then it points to nullptr.

```
struct Node {  
    int data;  
    Node* next;  
    Node* prev;  
};
```

1) Write a function called `remove_first()` that removes the first node from the list (where H is the head pointer). It should do nothing if the list is empty. Make sure there are no memory errors, and that the resulting list remains the same (just with the first node removed).

```
void remove_first() {  
    if (H == nullptr) return;  
    Node* p = H;  
    H = H->next;  
    if (H != nullptr) {  
        H->prev = nullptr;  
    }  
    delete p;  
}
```

2) Write a function called `inc_all()` that modifies the list by adding 1 to every node value. It should work for a list of any size.

```
void inc_all() {  
    Node* p = H;  
    while (p != nullptr) {  
        p->data++;  
        p = p->next;  
    }  
}
```

3)

- a. State the mathematical definition of “ $f(n)$ is $O(g(n))$ ” as given in the textbook.

$f(n)$ is $O(g(n))$ if there is a real constant $c > 0$ and an integer constant $n_0 \geq 1$ such that:

$$f(n) \leq cg(n), \text{ for } n \geq n_0$$

1 mark for real constant $c > 0$

1 mark for integer constant $n_0 \geq 1$

2 marks for $f(n) \leq cg(n)$

1 mark for $n \geq n_0$

- b. What is the **tightest** O-expression bound for $\log_2(n \cdot 8^n)$? Write your answer in the form $O(f(n))$, for some function $f(n)$, and show how you derived it.

The answer is $O(n)$ because:

$$\log_2(n \cdot 8^n) = \log_2 n + \log_2 8^n = \log_2 n + \log_2 2^{3n} = \log_2 n + 3n$$

Since $3n$ dominates $\log_2 n$, the expression is $O(n)$.